SC3102: Signals, Systems, and Transforms

Tutorial 04: Difference equations and Laplace transforms

Questions 1–2 concern real discrete-time signals and LCCDEs.

1. Consider the following LCCDE:

$$y[n] + \tfrac{1}{4}y[n-2] = 3\cos\left(\frac{\pi n}{2}\right)u[n], \qquad n \geq 0,$$

with initial conditions $y[-1] = 0$ and $y[-2] = 4$.

(a) Use the unilateral Z-transform to find the zero-input, zero-state and complete responses. *Hint:* Use partial fractions in z^2 and the cosine pair.

(b) Identify the transient and steady-state components.

(c) Can you set the initial conditions to eliminate the transient component? If so, find them.

2. Suppose a discrete-time LTI system satisfies

$$y[n] - y[n-1] - \tfrac{3}{4}y[n-2] = x[n-k], \qquad n \in \mathbb{Z}.$$

Answer the following for $k = 2$ and $k = -1$.

(a) Use the bilateral Z-transform to find $H(z)$.

(b) Make the pole-zero plot of $H(z)$ and indicate all its possible ROCs.

(c) Which of these systems are causal? Which are BIBO stable?

(d) Can any of these systems be causal **and** BIBO-stable?

(e) Find the causal impulse response if one exists.

Questions 3–4 concern continuous-time signals and Laplace transforms.

3. Invert the following Laplace transforms using the partial fraction method:

(a) $A(s) = \dfrac{s+5}{(s+2)^2}, \qquad \Re s > -2.$

(b) $B(s) = \dfrac{5}{(s+1)(s^2+4)}, \qquad \Re s > 0.$

(c) $C(s) = \dfrac{5}{(s+3)(s^2+2s+2)}, \qquad \Re s > -1.$

Hint: For the above two signals, use the partial fraction expansion and determine P, Q, R:

$$\frac{5}{(s+a)(s^2+cs+d)} = \frac{P}{s+a} + \frac{Qs+R}{s^2+cs+d}.$$

4. Consider the Laplace transform expression

$$F_a(s) = \frac{s-a}{(s+1)(s-2)}, \qquad a \in \mathbb{R}.$$

(a) For $a = 0$, make the pole-zero plot and sketch the possible ROCs.

(b) For $a = 0$, find $f_0(t)$ for each ROC. Which of them has a defined CTFT?

(c) For which values of a does the right-sided inverse have a CTFT? Sketch the corresponding pole-zero plot and ROC.

Useful Properties and Inverse Pairs

Z-transform:

$x[n]$	Z-transform $X(z)$	ROC
$p^n u[n]$	$\dfrac{z}{z-p} = \dfrac{1}{1-pz^{-1}}$	$\lvert z\rvert > \lvert p\rvert$
$-p^n u[-n-1]$	$\dfrac{z}{z-p}$	$0 < \lvert z\rvert < \lvert p\rvert$
$\rho^n \cos(\Omega_0 n)u[n]$	$\dfrac{z(z-\rho\cos\Omega_0)}{z^2 - 2\rho z\cos\Omega_0 + \rho^2}$	$\lvert z\rvert > \rho$
$\rho^n \sin(\Omega_0 n)u[n]$	$\dfrac{\rho z\sin\Omega_0}{z^2 - 2\rho z\cos\Omega_0 + \rho^2}$	$\lvert z\rvert > \rho$

Unilateral Shift Rules:

$$\mathcal{Z}_+\{x[n-1]\} = z^{-1}X_+(z) + x[-1],$$
$$\mathcal{Z}_+\{x[n-2]\} = z^{-2}X_+(z) + x[-1]z^{-1} + x[-2].$$

Bilateral Shift Rule:

$$x[n-k] \longleftrightarrow z^{-k}X(z), \qquad k \in \mathbb{Z}.$$

For both the above shift rules, the ROC is unchanged.

Laplace Transform Pairs:

Signal $x(t)$	Transform $X(s)$	ROC
$e^{pt}u(t)$	$\dfrac{1}{s-p}$	$\Re s > \Re p$
$-e^{pt}u(-t)$	$\dfrac{1}{s-p}$	$\Re s < \Re p$
$te^{pt}u(t)$	$\dfrac{1}{(s-p)^2}$	$\Re s > \Re p$
$\cos(\omega_0 t)u(t)$	$\dfrac{s}{s^2+\omega_0^2}$	$\Re s > 0$
$\sin(\omega_0 t)u(t)$	$\dfrac{\omega_0}{s^2+\omega_0^2}$	$\Re s > 0$
$e^{\alpha t}\cos(\omega_0 t)u(t)$	$\dfrac{s-\alpha}{(s-\alpha)^2+\omega_0^2}$	$\Re s > \alpha$
$e^{\alpha t}\sin(\omega_0 t)u(t)$	$\dfrac{\omega_0}{(s-\alpha)^2+\omega_0^2}$	$\Re s > \alpha$